

UNIT 5



COMPUTER SECURITY AND ETHICS



Q.1: WHY IS COMPUTER SECURITY IMPORTANT? WRITE ANY THREE REASONS.**IMPORTANCE OF COMPUTER SECURITY:****1. Prevent from viruses and malware:**

- Computer security is important for our computer's overall health.
- It keeps our information protected and helps prevent viruses and malware, which allows programs to run quicker and smoother.

2. Secure Private Information:

- It safeguards confidential and sensitive information. The advancement in science and technology has changed day to day activities. We rely on computers and mobile phones, where we access a lot of information which we do not want to share with others. This information may include our passwords, banking details, contacts, pictures, etc. To protect this information we need to make our devices secured that no one can damage or access this information without our consent.

3. Provide Safe Environment:

- Computer Security is important as it enables people to perform their work in safe environments.
- It helps in the execution of essential business processes.

Q.2 WHAT IS CREDIT AND DEBIT CARD SCAM?**(II) CREDIT AND DEBIT CARD SCAM:**

Keeping debit or credit cards is a common practice but insecure use of these cards can be dangerous. If a person has information about our debit or credit card he or she can make fraudulent transactions. There are various ways to get this information. One way is through scamming. Scammers set small machines inside an ATM or credit card machine. These machines copy the data which is then misused by the scammers. Debit and credit cards are also secured with PIN codes. User has to keep this code secret otherwise any person can use the card for online shopping and other purposes. All he or she needs to know our credit card number, PIN and security code printed on the back of the cards.

Q.3: WHAT DO YOU KNOW ABOUT COMPUTER SECURITY?**COMPUTER SECURITY:**

"Computer security is the protection against theft or damage to our computer hardware, software and information present on them from threat of viruses or unauthorized use."

Explanation:

The computer has become an important part of our life. We store important data on our computers in the shape of documents, pictures, programs, etc. Therefore, we expect that all our information must remain safe and our computer runs properly without any problem. Few threats can cause problems for our computers. These threats may be different types of viruses or unauthorized use of a computer. To prevent our computer from such threats, we need to abide by computer security.

Q.4: WRITE A SHORT NOTE ON CYBERCRIME.**CYBERCRIME:**

As communication, trade, and services are relying more on computers and networks. cybercrimes are growing too. Cybercrime is the crime that is committed through a computer and network. Cybercriminal uses devices to gain unauthorized access to important information. As Stealing passwords and important information, hacking social media accounts, accessing any individual bank account and making transactions, committing online frauds are some of the examples of cybercrime. Cybercrime is illegal and also punishable. According to Pakistan's Cybercrime Law, any offender who interrupts the privacy of a person or organization and harms their reputation may be sent to jail for three to five years including a heavy fine.

Q.5: WHO IS A CRACKER?**CRACKERS:**

Crackers are who gain unauthorized access to another system. They bypass passwords or licenses of computer programs, change source code or intentionally breach computer security. They do it with negative intentions. Crackers can also make targeted system unavailable or non-functional. They commit these activities generally for money but they may do it for fame or just for challenge or fun.

Q.6: WRITE DOWN SOME SAFEGUARDS AGAINST MALWARE.**SAFEGUARD AGAINST MALWARE:**

Keeping ourselves safe from malware and viruses is mostly in our hands. More than 90% of computers are infected due to the user's mistake. Our computers have caught a virus if they start slowing down, behave unusually, crash during processes or restart several times, show annoying messages and some of our documents disappear or become inaccessible. We must avoid this situation to be created. Some simple measures can prevent our system from malware and viruses. Install anti-virus software and keep it updated. Run scheduled scans regularly with any anti-virus software.

Keep our operating system updated.

- Do not click on internet links which have unusual labels, images or captions.
- Do not open email attachments or click on hyperlinks from unknown senders.
- Scan USB flash drive, SD cards and mobile phones before opening.
- Use spam blocking or filtering tools to block unsolicited emails, instant messages and pop-ups.

Only download files and programs from trusted source on the Internet, Never use an open Wi-Fi.

Q.7: DEFINE AUTHENTICATION MECHANISM.**AUTHENTICATION MECHANISM:**

The authentication mechanism is the hardware or software-based mechanic that forces users to prove their identity before accessing data on a device. The process makes sure the only authenticated user gets access to data or devices.

Q.8: DEFINE MALWARE.**MALWARE:**

The term malware is the contraction of malicious software. Malware is a broad term that encompasses computer viruses, worms, spyware, aware and others. Malware is a program that is written generally to cause a mess. They can be so dangerous that they can also damage devices. However commonly malware encrypt, steal or delete data, hijack core functions of computing and disturb different activities.